

The End of ASIC Proof-of-Work

The Security Condition

Bitcoin's Proof-of-Work security operates through a single mechanism: attacking the network requires controlling more computational power than all honest participants combined. The cost of acquiring and sustaining that power — in hardware and electricity — must exceed the potential reward for the mechanism to hold.

This mechanism functions only when one condition is satisfied:

Mining is the highest-value use of the electricity and hardware it consumes.

When this condition holds, the operators of mining hardware have stronger economic reasons to protect the network than to attack it. Security and self-interest are aligned. When this condition fails, the alignment breaks.

This condition has failed.

The Cause: Permanent Electricity Competition

The economics of electricity competition between AI infrastructure and Bitcoin mining are quantifiable across price scenarios. AI compute infrastructure — accelerators, high-bandwidth memory, and the interconnect systems that integrate them — generates revenue per kilowatt-hour that is largely independent of Bitcoin's price. Current-generation accelerator infrastructure, drawing approximately one kilowatt of power per compute unit including server overhead, generates \$2.00 to \$4.00 in revenue per kilowatt-hour.

Mining revenue per kilowatt-hour, by contrast, moves directly with Bitcoin's price and network difficulty. At Bitcoin's approximate peak price of \$100,000, the network's average mining hardware generated roughly \$0.08 per kilowatt-hour in gross revenue. The most efficient hardware deployed at that price level reached approximately \$0.24 per kilowatt-hour. At current prices near \$67,000, those figures fall to approximately \$0.05 and \$0.16 per kilowatt-hour respectively.

The differential is structural across all realistic price scenarios:

At Bitcoin \$100,000 — near its historical peak — average hardware produced revenue of \$0.08/kWh against AI's \$2-4/kWh: a differential of 25 to 50 times. Best-in-class hardware at that price reached \$0.24/kWh: a differential of 8 to 17 times. At current prices the gaps are wider. At any Bitcoin price that has historically existed, the differential remains an order of magnitude.

The gap is not a function of Bitcoin's current weakness. It is a function of the structural difference in value production between the two technologies. Mining's revenue per kilowatt-hour scales with Bitcoin's price. AI inference revenue per kilowatt-hour does not. For mining to reach parity with AI infrastructure at the lower end of current

AI pricing, Bitcoin would need to sustain prices above \$1,000,000 per coin — a level at which the mining economics argument is the least of any holder’s concerns.

The consequence is that industrial electricity — the resource on which Bitcoin’s security model depends — now has a permanently higher-value competing use across all realistic market conditions. Electricity markets respond to higher bidders. Mining cannot match the prices AI operators pay. The pool of electricity accessible to miners at economics compatible with profitable operation has permanently contracted.

The foundational assumption of Proof-of-Work — that mining will always represent the optimal use of specialized hardware — has been falsified by the emergence of a technology that competes for the same resource and generates substantially more value from it across every price scenario that has historically applied to Bitcoin.

The Energy Refuge: An Insufficient Defense

A counterargument exists: remote regions with genuinely cheap electricity — stranded hydroelectric power, geothermal sources, flared gas — represent locations where AI cannot economically deploy, and where mining therefore remains the optimal use of available energy. If such locations continue to support mining operations, does the security model retain a foundation?

The argument misidentifies the nature of the security requirement. The question is not whether mining remains the optimal use of electricity in some locations. The question is whether the hashrate generated by those locations is sufficient to defend the network against attack.

The scale of genuinely AI-inaccessible mining is quantifiable. The Cambridge Digital Mining Industry Report published in April 2025 estimates that stranded natural gas supplies 507 megawatts of mining capacity, representing 3.3% of the industry’s total energy mix. Remote hydroelectric operations — Paraguay at approximately 4.3% of global hashrate, Laos at approximately 0.9%, and comparable smaller jurisdictions — account for an additional 3 to 5% of global hashrate. The United States, China, and Russia together control approximately 68% of global hashrate, all in locations with the infrastructure to support AI data center deployment. The total hashrate in locations that AI infrastructure genuinely cannot reach is approximately 5 to 8% of the global network. This fraction cannot defend a network against an attack requiring only 30 to 40% of total hashrate to produce visible disruption.

The global hashrate is dominated by large-scale industrial mining operations — facilities with dedicated power infrastructure, professional management, and the capital to secure favorable long-term electricity contracts. These operations represent the overwhelming majority of the network’s computational security. They are also the operations whose infrastructure is most compatible with AI hosting requirements, and therefore the operations most capable of completing the transition.

Once these large-scale operations transition, the hashrate they represented leaves the network. What remains in remote locations is insufficient in scale to maintain the network’s security threshold. The hardware retired by transitioning miners does

not disappear. Operations that have completed the transition to AI may retain their ASIC hardware — it remains available for use or sale. Those that sell require only a price above scrap value to transact, a threshold easily met by any actor assembling attack capacity. Failed-to-transition miners hold their own hardware under conditions of mounting financial distress. Both populations contribute to a pool of deployable attack hardware that grows as the industry transition accelerates.

The attacks that follow do not require competing for cheap electricity in remote regions. They require only that the cost of running existing hardware during the attack window is less than the profit generated by the short position. That condition holds regardless of what electricity costs in locations AI has not reached.

The price decline caused by attacks then eliminates the economic basis for even the most favorably situated remote mining. Mining revenue is denominated in Bitcoin. As Bitcoin's price falls in response to attacks and to the broader exit dynamics, the revenue from each block mined falls proportionally. An operation that was profitable at low electricity cost becomes unprofitable when the Bitcoin it produces is worth less. The refuge that appeared to exist because of cheap electricity is eliminated not by AI competition for that electricity, but by the economic consequences of the attacks that large-miner transition made possible.

The small remote miners do not constitute the network's last line of defense. They are the last to leave it.

The Supply Chain: A Ratchet With One Direction

New ASIC hardware requires sub-5-nanometer chip fabrication, available only at TSMC and Samsung. Both operate at capacity. AI chip production generates higher margins and represents higher strategic value to foundries than ASIC orders. Mining hardware manufacturers are losing foundry allocation priority.

Maintaining a foundry allocation requires continuously growing order volume. Manufacturers who cannot sell existing inventory cannot place larger future orders. Without larger orders, they lose their allocation permanently. Lost foundry position means lost manufacturing capacity. This process does not pause and wait for market conditions to improve. It is terminal.

The result: the directional shift is established. Foundry capacity that previously supported the expansion of Bitcoin's hashrate is being reallocated toward AI accelerator production at a pace the mining hardware market cannot absorb. The commercial pipeline contracts as the economics that sustained it deteriorate. Existing machines are a depleting stock without a replacement cycle operating at the scale the network requires.

The Secondary Market: Effective Cessation

The collapse of new hardware demand has been accompanied by the effective cessation of the secondary market for used ASIC equipment. This is a distinct and consequential development.

Hardware that previously held a recoverable fraction of its acquisition cost now trades, when it trades at all, at prices representing a small percentage of original value. Fixed-price secondary markets have been replaced by auction mechanisms in which sellers set no floor and buyers name their own price. The absence of a clearing price is not a feature of a depressed market. It is a feature of a market in which buyers have concluded that the asset has no reliable future value and therefore no present value worth committing capital to.

Manufacturers have responded by operating unsold inventory themselves. This practice is described as confidence in mining economics. It is more accurately described as a balance sheet decision: self-mining generates electricity revenue that partially offsets carrying costs, and this outcome is preferable to recording immediate inventory write-downs. The machines run not because mining is profitable but because the accounting treatment of running them is less damaging than the alternative.

The secondary market exit — selling hardware to recover capital — has ceased to function as a viable option. The one path that would allow capital recovery without attacking the network has been removed by the market itself.

Hashrate: The Peak Is Set

Bitcoin's network hashrate reached its historical maximum in late 2025. It has not broken that record since. In the period following, downward difficulty adjustments have outnumbered upward adjustments.

For hashrate to reach new historical highs, the following conditions would need to hold simultaneously: new machines deployed at scale, a production pipeline to supply them, and electricity at prices below the threshold set by AI operator demand.

None of these conditions are present. The ceiling has been established. The descent is not cyclical — it reflects the absence of the structural inputs that drove prior growth. Existing hardware ages and is retired. Marginal operations shut down. The difficulty adjustment compensates for reduced hashrate but does not reverse it. The stock of operating hardware declines monotonically.

The Weaponization of Stranded Hardware

Mining hardware that cannot operate profitably has three available uses: continued operation at a loss, sale at near-zero recovery value, or repurposing as an attack instrument.

The attack mechanism functions as follows. A short position is established in Bitcoin's derivatives markets — through centralized exchanges, decentralized protocols, or any venue where the position can be built and held. Mining hardware is then directed at the network to produce visible disruption — empty blocks, chain reorganization attempts, detectable anomalies sufficient to trigger exchange withdrawal suspensions and market panic. The attack does not require technical success. It requires only that the disruption be visible enough to produce a rapid price decline. The short position captures that decline at leverage.

The cost of operating hardware for 48 hours is finite. The potential return from a leveraged short position on a 30% price decline is orders of magnitude larger. The legal risk of this mechanism warrants examination, because it is not what it appears.

A 51% attack consists of miners directing their own hardware to mine valid blocks. No law in any jurisdiction prohibits a mining operation from possessing more than 50% of network hashrate. The act of mining empty blocks or executing a chain reorganization is the use of hardware that the operator legally owns, producing outcomes that the protocol itself permits. Shorting Bitcoin is legal trading activity. The combination of the two — directing hashrate to produce network disruption while holding a short position — is where the legal question would arise. But that question has no answer in any existing legal framework.

Bitcoin has no issuer. It has no board of directors. It has no regulatory subject who can be summoned before a court. A prosecution for market manipulation requires a jurisdiction with authority over the manipulated asset, a legal framework defining the manipulation, an enforcement mechanism capable of reaching the defendants, and an identifiable victim who can bring a claim. For Bitcoin, all four are simultaneously absent. The miners most likely to execute this mechanism operate in Russia, Kazakhstan, and Iran — jurisdictions with minimal judicial cooperation with Western regulatory bodies. The hardware is their property. The short position is legal. The protocol permits the mining behavior. There is no defendant, no statute, and no court.

Traditional market manipulation carries legal deterrence because regulated markets have regulatory bodies, securities laws define manipulation, and enforcement agencies have jurisdiction over market participants. None of these conditions apply to a decentralized protocol with no issuing authority and no geographic nexus. The legal risk is not a cost to be weighed against expected return. It is structurally absent.

The critical distinction is this: the attack does not need to be a single decisive event. Once network hashrate falls to the threshold at which an attack is economically executable, that condition does not reverse. The threshold itself is not fixed — it moves continuously as total network hashrate declines, making attacks progressively easier to execute in absolute terms. The condition of vulnerability, once entered, compounds and persists. Any actor who can assemble sufficient hardware can initiate a disruption at any point thereafter, repeatedly, for as long as the economic logic holds.

This transforms the nature of the risk from a one-time crisis into a permanent state. A network in which attacks are periodically executable and periodically profitable cannot provide the property its existence depends on: transaction finality. Every confirmation becomes conditionally reversible. Every exchange must treat withdrawal requests as potentially fraudulent during disruption events. Every large transaction

carries settlement risk that cannot be priced or insured against. The network does not fail in a single moment. It enters a condition of indefinite instability — functional in appearance during quiet intervals, unreliable in operation whenever the attack threshold is crossed.

The same machines that constituted the network's security become, under these conditions, the permanent source of its instability.

The Most Probable Attack Vector

The actors most likely to execute this mechanism are not the largest mining pools and not state actors. They are mid-size mining operations that attempted and failed to complete the transition to AI infrastructure.

Their situation has a specific structure. AI compute infrastructure — accelerators, high-bandwidth memory, and the integrated systems that combine them — requires capital that mining revenue, at current economics, cannot generate. The transition window — the interval during which a mining company could sell Bitcoin holdings and the capital cost of frontier AI compute remained reachable — has been closing as Bitcoin's price has declined and successive generations of AI compute infrastructure have commanded increasing capital requirements simultaneously. An operation that hesitated, or that lacked sufficient capital, finds itself in a position with no remaining exit.

Mining at a loss produces a known, ongoing cash drain. The ASIC hardware on the balance sheet carries a cost basis that the secondary market will not recover — the machines are worth a fraction of their acquisition price, and the market for them has largely ceased to function. The operation faces losses from both directions: the income statement and the asset base are deteriorating in parallel, with no business path that reverses either.

For these operators, the attack-plus-short mechanism is not an aggressive act. It is the only remaining mechanism by which the sunk capital in ASIC hardware can generate any return before the machines become worthless through attrition. Every month of continued mining deepens the loss. The attack represents the conversion of a depreciating, illiquid asset into a single, time-limited profit opportunity.

A question arises: faced with terminal financial distress, might these operators simply accept bankruptcy quietly rather than act? For an individual actor, passive dissolution is possible. For an industry, it is not how endings work.

No industry in the historical record has disappeared silently. Industries facing existential pressure lobby, organize, litigate, and search for exits. When legal and competitive exits are closed, the search continues toward whatever remains available. The mining industry has the organizational infrastructure to channel this process: years of established conferences, trade associations, equipment supplier networks, and direct communication channels across the major operating jurisdictions. Shared desperation does not need to create new coordination structures. It travels through the ones that already exist.

When the mechanism available at the end of that search is the attack-plus-short trade — and when, as established above, the legal risk of executing it is structurally absent — the collective logic of a financially terminal industry converges there. Individual operators may hesitate. The industry, as a collective, will not.

The scale of this population matters. When the actors are the majority of an industry rather than a minority, the combined hashrate available for a coordinated action is not a marginal fraction of the network — it is the network’s remaining base. Pool concentration in the current network further lowers the number of actors whose coordination is required to produce visible disruption. The question is not whether sufficient hashrate exists among stranded operations to execute the mechanism. The question is when the financial pressure on that population reaches the point where acting together is more rational than continuing to lose money separately.

The Miner Transition

Mining facility infrastructure — power delivery systems, cooling capacity, network connectivity — is directly compatible with AI data center requirements. The transition from mining to AI hosting does not require construction from zero. It requires replacing one category of hardware within existing facilities.

AI hosting contracts offer predictable, dollar-denominated revenue under long-term service agreements. Mining offers volatile revenue exposed to coin price, difficulty, and halving cycles, with no contractual floor.

Capital allocates toward predictability. The mining industry is transitioning. Listed mining companies are selling Bitcoin to fund AI compute infrastructure, signing AI hosting contracts, and explicitly stating in regulatory filings that capital allocation to mining cannot be justified at current economics.

Once power capacity is committed to long-term AI leases, it is no longer available for mining. The transition does not reverse when mining conditions improve. The infrastructure is permanently reallocated.

The operators of the hardware that secures Bitcoin’s network are rationally and systematically exiting the role of securing it.

The Demand Scale

Bitcoin’s global mining network draws approximately 15 to 20 gigawatts of continuous power. United States AI data center capacity additions are projected at 13.6 gigawatts in 2026 and 36.3 gigawatts in 2027. Global AI workload demand is projected to reach 156 gigawatts by 2030. The mining network’s total power draw is smaller than a single year of incremental AI capacity additions in the United States alone.

The primary constraint on AI data center development is not capital. It is not demand. It is access to committed industrial power at sufficient scale. New grid interconnec-

tion requests in established U.S. markets carry queue times extending beyond five years. A large mining facility represents exactly the asset class that AI operators cannot obtain through conventional development on any near-term timeline: committed grid connections at industrial scale, with existing high-voltage infrastructure already in place and operational.

The geographic alignment is structural. The United States operates 37.8% of global hashrate, with Texas representing its largest single concentration. Texas is simultaneously the leading destination for AI data center development, projected to exceed 40 gigawatts of data center capacity by 2028. The power resources that support Bitcoin's largest national hashrate concentration are the resources that AI infrastructure development is most actively acquiring. The overlap is not coincidental. Both industries are drawn to the same underlying asset.

The transition is not commercially constrained. The demand exists at a scale that dwarfs the available supply. The transition is commercially compelled.

The conversion of a mining facility to AI hosting requires fiber infrastructure, liquid cooling systems, and power distribution upgrades. The process requires 18 to 36 months under favorable conditions. During this interval, the operator has committed capital to the transition while mining revenue continues to deteriorate and AI revenue has not yet come online. The exposure runs in both directions simultaneously.

This interval is not an obstacle to the transition. It is the period in which the economic conditions described in the preceding sections reach their most acute form.

The Protocol Cannot Repair This

Each available adaptation pathway either destroys the condition it is meant to preserve, or fails to address the root cause.

Changing the mining algorithm retires all existing hardware instantly — eliminating the network's entire computational security at the moment the transition window is open and exposure is maximum.

Transitioning to Proof-of-Stake introduces a problem larger than the one it is meant to solve. In a Proof-of-Stake system, network control is proportional to staked holdings. A significant fraction of Bitcoin supply is permanently dormant — lost keys, long-term cold storage, untouched early holdings. The actively staked portion of supply would represent a subset of total supply. Within that subset, the largest institutional holders would constitute a disproportionate share of effective network control.

The transition would hand dominant influence over the network to the entities that accumulated the most Bitcoin through leveraged institutional operations. The network designed as an alternative to institutional financial control would become controlled by institutions. The fix produces an outcome that contradicts the system's foundational premise more completely than the problem it was meant to address.

Additionally, the transition window itself remains exposed. During the interval in which the old security mechanism is being dismantled and the new one is not yet

operational, holders of existing mining hardware have inverted incentives: attacking the network in transition is more economically rational than participating in it.

Modifying the fixed supply cap — adding ongoing issuance to sustain mining rewards — does not address the root cause and therefore does not constitute a solution.

The problem is not that mining rewards are insufficient in absolute terms. The problem is that the revenue generated per kilowatt-hour of electricity consumed by mining is structurally below the revenue generated by competing uses of that electricity — at every Bitcoin price that has historically existed. As established above, even at Bitcoin’s peak price of \$100,000, best-in-class mining hardware reached approximately \$0.24 per kilowatt-hour against AI’s \$2 to \$4 per kilowatt-hour. The differential persists across price scenarios because mining revenue scales with Bitcoin’s price while AI inference revenue does not. This differential exists at the level of energy economics, not monetary policy.

Adding block rewards changes the quantity of Bitcoin issued. It does not change the per-kilowatt-hour value of mining relative to AI. For mining to recover competitiveness in electricity markets, the revenue per kilowatt-hour from mining would need to rise by an order of magnitude — requiring not a marginal adjustment to issuance but a combination of price and emission levels that would represent a fundamental restructuring of the monetary system. Supply modification without closing the per-kilowatt differential is inert against the underlying cause.

The protocol’s strength and its vulnerability are the same attribute. Immutability is simultaneously the source of Bitcoin’s value proposition and the barrier to the adaptation that value proposition now requires. Each modification pathway either announces the elimination of the property being protected, produces an outcome worse than the problem it addresses, or fails to operate at the layer where the problem exists.

The Three Conditions of PoW Security

Proof-of-Work security requires three conditions to hold simultaneously.

First: sufficient computational power that controlling a majority is prohibitively expensive.

Second: the operators of that computational power have stronger economic reasons to protect the network than to attack it.

Third: no mechanism exists by which attacking the network generates a return superior to honest mining.

First condition: hashrate is in structural decline with no production pipeline to reverse it.

Second condition: miners are transitioning to AI infrastructure. Those who cannot transition face economic logic that points toward the attack mechanism as the only remaining path to capital recovery.

Third condition: the attack-plus-short-position trade generates returns orders of magnitude larger than honest mining under current and projected conditions.

All three conditions are simultaneously absent.

The Nature of a Stable Causal Chain

A causal chain is stable when each link follows necessarily from the previous one — not through coincidence, not through the accumulation of errors, not through the actions of any identifiable antagonist, but through the rational responses of individual actors to the conditions they face. When every step in a sequence is individually rational, the sequence does not require external force to complete. It completes because each participant, acting in their own interest, advances it.

The chain described in this document has that structure at every link. Miners transition to AI because AI contracts generate higher and more predictable returns than mining. Those who cannot transition face an economic calculus in which attacking the network is more rational than continuing to operate at a loss. The protocol cannot adapt because every adaptation pathway destroys the source of value it is trying to preserve. Each link follows from the one before it by the ordinary logic of self-interest operating within changed conditions.

The historical record of stable causal chains is instructive not as analogy but as pattern.

Rome's administrative and military costs began exceeding the revenue that imperial scale could generate during the Crisis of the Third Century. The Western Empire fell in 476 AD. The chain from structural failure to completion required approximately **200 years**. No internal reform could resolve the contradiction without dismantling the imperial structure itself. The chain completed.

The Soviet economic model presents a more precisely measurable case. In 1985, Mikhail Gorbachev assumed leadership and immediately launched Glasnost and Perestroika — programs explicitly designed to reform a system its own leadership had concluded could no longer function as constituted. This is the moment the structural failure was acknowledged at the highest level of the system itself. The Soviet Union dissolved in December **1991**. From official acknowledgment of structural failure to completion: **6 years**. No reform short of abandoning the system's central premise could address the underlying condition. The chain completed.

The contrast between these two cases is instructive. Rome required 200 years because information traveled at the speed of a courier and capital at the speed of a cart. Each step of the deterioration was separated from the next by intervals measured in decades. The Soviet case compressed to 6 years in a mid-twentieth century information environment — faster than Rome, but still bounded by the friction of that era's communication and financial infrastructure.

Bitcoin's structural failure became observable in its publicly verifiable data in 2025: hashrate peaked, difficulty adjustments turned persistently negative, mining economics fell below production costs across a significant portion of the network, and

the transition of mining infrastructure to AI hosting became a disclosed, reported phenomenon. This information is not available to analysts after a delay. It is visible to every participant simultaneously, in real time, updated with each block.

The interval between Rome's 200 years and the Soviet Union's 6 years reflects the difference in information infrastructure between those eras. The interval between the Soviet Union's 6 years and Bitcoin's transmission environment reflects a further compression of the same magnitude. The pattern that required 200 years to complete under Roman conditions, and 6 years under mid-twentieth century conditions, operates in an environment where difficulty adjustments are announced every two weeks, miner wallet flows are tracked by the hour, and exit from any position executes in seconds.

In each case, the causal chain — once the environmental condition changed — completed regardless of the intentions of those inside it.

There is a further observation that the historical cases make available. The Romans who believed Rome was permanent were not being irrational. Rome had existed for nearly a thousand years. Its laws, roads, and language had penetrated every corner of the known world. Every piece of visible evidence supported the conclusion that it would continue. The belief in permanence was the most reasonable inference available from the evidence at hand.

The Soviet engineers, scientists, and citizens who believed their system represented the direction of history were not being deceived in any simple sense. Within the information environment available to them, the conclusion that the system was structurally sound was a rational one. The failures were not visible from the inside until the transmission had already begun.

What this reveals is a specific property of systems in which a causal chain has become stable but has not yet completed: the apparent solidity of the system is real, but it is a product of past performance, not present structural integrity. The system continues to function. Institutions continue to operate. Participants continue to transact. The external evidence of stability is genuine. But the conditions that produced that stability have been removed, and what remains is the accumulated inertia of a system whose foundation is no longer present.

This is not collective irrationality. It is the rational application of historical evidence to a present in which that evidence no longer describes the underlying reality. The belief in the system's permanence is based on the fact of its past permanence. That inference fails precisely when the conditions that made permanence possible have changed.

In each case, the system was genuinely powerful, the belief in its permanence was reasonable given available evidence, and the core strength of the system became the specific barrier to the adaptation it required. The causal chain completed regardless of the intentions of those inside it.

The apparent solidity of a system whose underlying logic has failed is not strength. It is the interval between the failure of the foundation and the recognition of that failure. The interval can be long. The transmission, when it begins, tends to be

sudden — because what closes is not a gradual deterioration but the gap between the perceived state of the system and its actual structural condition.

This creates an information environment whose current state is structurally abnormal. When a major asset is in genuine uncertainty, market sentiment is divided. Serious analysis exists on both sides. The current public discourse on Bitcoin has a different shape: near-unanimous confidence in a cyclical interpretation — excessive leverage, historical patterns pointing to a bottom, network security intact — at precisely the moment when every relevant data signal points in the opposite direction. Hashrate declining. Difficulty adjustments persistently negative. Miners systematically selling. Each of these data points is individually visible. Their structural interpretation is almost entirely absent from mainstream discourse.

The mechanism that produces this abnormality is not conspiracy. Every institution holding Bitcoin cannot acknowledge the structural failure without accelerating its own losses before its exit is complete. Every mining company cannot acknowledge it without collapsing the equity value that funds operations. Every exchange operator cannot acknowledge it without losing users. Every media outlet covering Bitcoin cannot acknowledge it without losing its audience. Each of these actors, independently, arrives at the same rational conclusion: maintain the prevailing narrative while managing their own position. No coordinating meeting is required. No agreement among parties exists. There is only a set of interest structures producing identical behavior across an entire ecosystem — an ecosystem in which selling is occurring while the narrative that attracts buyers is simultaneously being maintained.

This structure is more difficult to address than deliberate manipulation. A conspiracy can be investigated, exposed, and prosecuted. Structural self-interest has no responsible party, no document that reveals it, and no intervention point that terminates it. It is the normal operation of a system whose participants are responding rationally to their circumstances.

The closest structural parallel is the period before the 2008 subprime crisis. The information asymmetry had the same shape: a small number of analysts had assembled the complete picture; the mainstream narrative was based on genuine historical evidence that had ceased to describe current structural reality; the interest structure imposed the same silence across rating agencies, investment banks, and financial media; and a short mechanism existed that gave those who understood the risk every reason to build positions quietly rather than speak publicly. The pattern of informed exit occurring simultaneously with narrative maintenance — ensuring the liquidity that exit requires — was identical.

The chain does not require anyone to intend the outcome. It requires only that the conditions exist and that actors respond to those conditions rationally. Both are true here.

The Compound Mechanism: Reflexive Spiral and Instant Transmission

The structural failures described in this document do not operate as independent, sequential events. They interact through a feedback loop in which each step gen-

erates information, that information is transmitted instantly to all participants, and the responses those participants take advance the next step. The reflexive spiral and the speed of transmission are not two separate risks. They are a single compound mechanism whose destructive potential is determined by their interaction.

The spiral operates as follows.

Mining companies that recognize the transition window sell Bitcoin to fund AI compute infrastructure. This selling pressure depresses Bitcoin's price. A lower Bitcoin price simultaneously reduces mining revenue and increases the relative cost of AI compute infrastructure — meaning the capital available for transition shrinks while the cost of transition rises. Operations that hesitated find the window has closed. They are stranded with ASIC hardware whose secondary market value is near zero, mining at a loss with no viable path forward. The hashrate they represent remains on the network, deteriorating the economics for every remaining miner. Further miners exit. Hashrate falls. The attack threshold becomes easier to reach. An attack occurs. Price falls sharply. The remaining mining operations shut down or complete their exit. The cycle has advanced.

Each step of this sequence generates observable data. Bitcoin wallet flows are public. Mining difficulty adjustments are announced every two weeks and quantify the exact rate of hashrate loss. Corporate filings disclose miner Bitcoin sales. Derivatives markets show the accumulation of short positions in real time. Exchange withdrawal volumes are tracked. There is no point in this sequence at which the information that drives the next response is unavailable to participants.

The significance of this is not simply that information is public. It is that instant, universal information access eliminates the intervals between spiral steps that would otherwise allow corrective action. In a prior era, the deterioration of a financial system's fundamentals could remain invisible to most participants for months or years. The delay created space: for regulators to intervene, for capital to find alternative support, for participants to gradually price in the change rather than responding simultaneously.

That space does not exist here. When a difficulty adjustment is announced, every mining operation in the world receives the same signal at the same moment. When a major corporate holder files a disclosure of Bitcoin sales, every market participant reads it simultaneously. When an exchange suspends withdrawals in response to a network disruption, the news reaches every holder before the suspension is lifted. The interval between signal and response has been reduced to the latency of a network connection.

The consequence is that the spiral does not advance gradually. Each step, once triggered, generates a signal that produces responses at the speed of information — and those responses constitute the next step. The interval between steps approaches zero. A process that in an earlier financial era might have represented a multi-year deterioration, during which participants retained the ability to observe, deliberate, and respond, becomes a process in which the steps follow each other faster than deliberate response is possible.

The exit mechanism amplifies this further. Converting a Bitcoin position to cash or

stable currency requires no intermediary, no business hours, no geographic proximity to a financial institution. It executes instantly, across every jurisdiction, continuously. When panic begins, every holder acts at the same time. There is no queue, no delay, no friction that staggers the responses and prevents simultaneous execution from collapsing liquidity.

The derivatives market introduces the final element. Short positions profit from the transmission of deterioration. The actors who hold them do not wait for the spiral to advance on its own schedule. They have financial incentives to accelerate each step — to make the information visible, to amplify the signal, to ensure that each piece of negative data reaches participants before any stabilizing narrative can form. The spiral is not merely transmitted by the information infrastructure. It is actively accelerated by participants whose returns scale with the speed and depth of each step's completion.

The compound mechanism produces an outcome qualitatively different from either component operating alone. A reflexive spiral without rapid transmission would deteriorate slowly enough to allow observation and response. Rapid transmission without a reflexive structure would move information quickly but without an underlying process to transmit. Together, they create a system in which the interval between the beginning of the spiral and its completion is compressed to the point where the corrective window — the time available between recognition and irreversibility — approaches zero.

The contrast between Rome's 200 years and the Soviet Union's 6 years reflects the difference in information infrastructure between those eras. Each step of Rome's deterioration was separated from the next by intervals long enough for responses to form. In the Soviet case, even with mid-twentieth century communication constraints, the interval compressed to 6 years once the structural failure was acknowledged. The intervals that separated Rome's steps no longer exist in the current environment. What required 200 years under Roman conditions can complete, under the compound mechanism described here, in the interval between a visible triggering event and the exhaustion of exchange liquidity.

Conclusion

The economic condition that made Proof-of-Work security viable — the alignment between miner self-interest and network protection — has been permanently disrupted by a technology that generates higher value from the same resource that mining requires.

The ASIC supply chain is contracting, with foundry allocation shifting toward AI accelerator production at a rate the mining hardware market cannot reverse. The secondary market for mining hardware has effectively ceased to function, removing the last capital recovery option for stranded operators outside of the attack mechanism. Hashrate has peaked and will not recover under present conditions. Stranded hardware is being converted from a security instrument into an attack instrument by financial logic, without requiring coordination or malicious intent. The protocol cannot

adapt without destroying what it is adapting.

The three structural conditions that define Proof-of-Work security are simultaneously absent.

Once network hashrate falls to the point at which an attack becomes economically rational, that condition does not reverse. The threshold itself moves continuously as hashrate declines, making attacks progressively easier to execute. The attack can be repeated by any actor with sufficient hardware, at any time, for as long as the underlying economic logic holds. The network does not require a single catastrophic failure to lose its utility. It requires only the persistent possibility of disruption — which prevents transaction finality from being reliable, which is the only property the network exists to provide.

Bitcoin's Proof-of-Work security model does not have a foundation on which it currently operates. The causal chain is complete. Each step has begun. What follows is not a crisis that resolves. It is a condition that persists.

Data referenced in this analysis is drawn from public filings, on-chain metrics, and market data as of June 2026.